

Chapter

4

Angular Motion

ROTATIONAL KINEMATICS:

Angular motion, or rotational kinematics, is the field of physics that focuses on describing how objects move when they spin around an axis. This branch of physics plays a crucial role in helping us understand the motion of objects such as wheels, gears, planets, and various other systems as they travel in circular or angular paths.

Key concepts in angular motion include angular displacement, angular velocity, and angular acceleration, which are analogous to their linear counterparts in translational motion.

ANGULAR DISPLACEMENT:

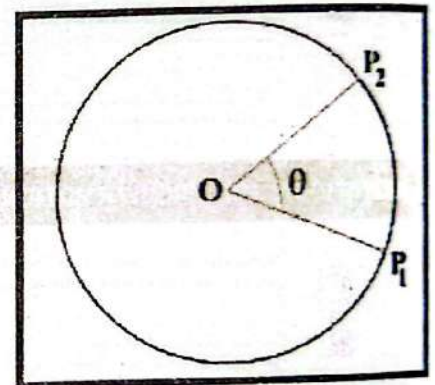
Angle subtended at the center by a body moving in a circular path is called its angular displacement.

Explanation

Consider a particle moves in a circular path from a point P_1 to P_2 in an interval of time Δt . It travels an arc. The angle $\angle P_1OP_2 = \theta$ which is called angular displacement of the particle

Where

$$\theta = \frac{s}{r}$$

**RADIAN**

"One radian is defined to be the angle subtended, where the arc length 'S' is exactly equal to the radius of the circle".

Relation between Degree and Radian

For one complete revolution $\theta = 360^\circ$ then 'S' becomes the circumference of circle
As we know for a complete rotation:

$$S = 2\pi r$$

$$r\theta = 2\pi r$$

$$\theta = 2\pi \text{ radians}$$

$$\theta = 360^\circ = 2\pi \text{ radians}$$

$$\text{Therefore 1 radian} = \frac{360}{2\pi} = 57.3^\circ$$

$$1 \text{ degree} = \frac{2\pi}{360} = 0.01745 \text{ rad}$$

ANGULAR VELOCITY:

Angle displacement in unit time is called angular velocity.

OR

The rate of change of angular position of a body is called angular velocity.

Explanation

Consider a particle moving in a circular path of radius "r".

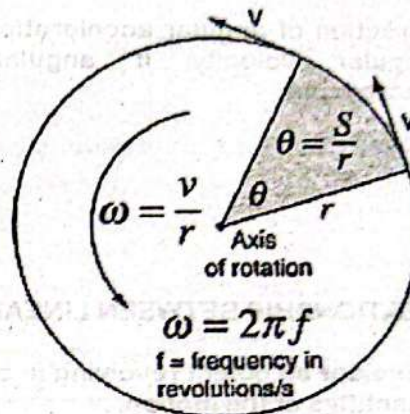
The angular displacement is $\theta_2 - \theta_1 = \Delta\theta$

Time interval = $t_2 - t_1 = \Delta t$

Since the angle traced per sec by a body moving in circle is called its average angular speed denoted by " ω_{av} ".

Therefore,

$$\omega_{av} = \frac{\Delta\theta}{\Delta t}$$

**Unit**

Unit for angular velocity is rad /sec OR rev/sec.

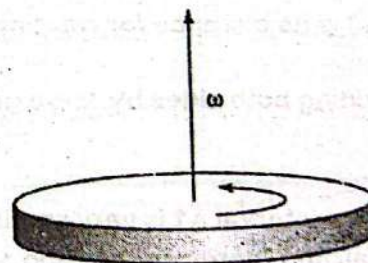
INSTANTANEOUS ANGULAR VELOCITY

The angular velocity at a particular instant is called instantaneous angular velocity.

If Δt is very small i.e. $\Delta t \rightarrow 0$ then $\Delta\theta / \Delta t$ is called instantaneous angular velocity.

Direction Of Angular Velocity

Direction of angular velocity is given by **RIGHT HAND RULE**, which states that if we curl the fingers of right hand in the direction of rotation then thumb will represent the direction of angular velocity.

**ANGULAR ACCELERATION:****Definition**

Rate of change of angular velocity of a body is called angular acceleration.

Explanation

It is denoted by α . If ω_1 is angular velocity of a body moving in circle at instant t_1 and ω_2 is angular velocity of body at instant t_2 then,

$$\alpha_{av} = \frac{\omega_2 - \omega_1}{t_2 - t_1}$$

$$\alpha_{av} = \frac{\Delta\omega}{\Delta t}$$

If Δt is very small then acceleration is called instantaneous angular acceleration.

$$\alpha_{ins} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\omega}{\Delta t}$$

Unit

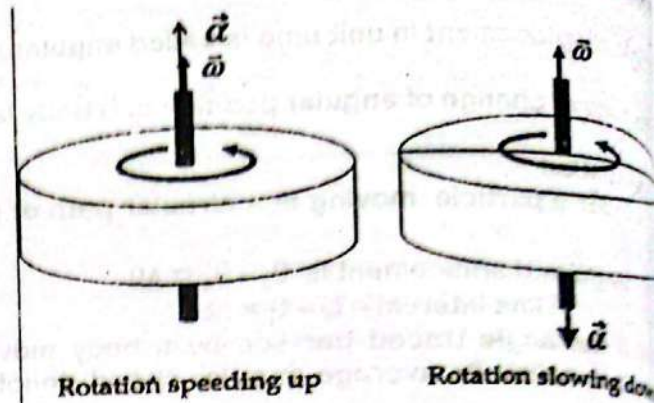
Unit for angular velocity is rad / sec².

Physics XI

Direction

Direction of angular acceleration is same as angular velocity if angular velocity is increasing.

Direction of angular acceleration is opposite to angular velocity if angular velocity is decreasing.



RELATIONSHIP BETWEEN LINEAR AND ANGULAR QUANTITIES:

Consider an object revolving in circle, In order to find the relations between linear and angular quantities of the motion.

We know that:

Relation Between Angular Distance And Linear Distance:

$$S = r\theta$$

$$\theta = S/r$$

Relation Between Angular Velocity And Linear Velocity:

If ΔS is its distance for rotating through angle $\Delta\theta$ then,

$$\Delta\theta = \Delta S/r$$

Dividing both sides by Δt we get:

$$\Delta\theta/\Delta t = (\Delta S/r) / \Delta t$$

$$r\Delta\theta/\Delta t = \Delta S/\Delta t$$

If time interval Δt is very small $\Delta t \rightarrow 0$, then the angle through which the particle moves is also very small and therefore the ratio $\Delta\theta/\Delta t$ gives the instantaneous angular speed ω_{ins} .

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta S}{\Delta t} = r \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} \quad \text{-----eq(1)}$$

Now by definition

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta S}{\Delta t}$$

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t}$$

Therefore equation (1) reduces to:

$$V = r\omega$$

RELATION BETWEEN ANGULAR ACCELERATION AND LINEAR ACCELERATION:

Suppose an object rotating about a fixed axis changes its angular velocity by $\Delta\omega$ in time Δt sec, then the change in tangential velocity ΔV at the end of this interval will be

$$\Delta V = r\Delta\omega$$

Change in velocity in unit time is given by:

$$\frac{\Delta V}{\Delta t} = r \cdot \frac{\Delta\omega}{\Delta t}$$

If Δt approaches to zero then $\Delta V / \Delta t$ will be instantaneous tangential acceleration and $\Delta\omega / \Delta t$ will be instantaneous angular acceleration " α ".

$$a = r\alpha$$

The Equation of motion can be written as:

EQUATIONS OF LINEAR MOTION	EQUATION OF ANGULAR MOTION
$S = Vt$	$\theta = \omega t$
$S = V_i t + \frac{1}{2}at^2$	$\theta = \omega_i t + \frac{1}{2}\alpha t^2$
$V_f = V_i + at$	$\omega_f = \omega_i + \alpha t$
$2aS = V_f^2 - V_i^2$	$2\alpha\theta = \omega_f^2 - \omega_i^2$

Chapter**4****Centripetal Acceleration****CENTRIPETAL ACCELERATION:**

"When an object moves in a circle, the magnitude of the velocity remains same, but the direction of velocity changes at every point during the circular motion. Due to changing the direction of velocity an acceleration is produced, which is always directed towards the center of the circle. It is called centripetal acceleration".

It is denoted by $\vec{a}_c$.

DERIVATION:

In order to calculate the magnitude of centripetal acceleration a_c we must first find the velocity difference ΔV for two successive positions of an object moving along a circular path.

Suppose the object takes a time $\Delta t = t_2 - t_1$ to go from position P_1 to position P_2 .

Physics XI

The vector difference ΔV is due to the different directions of the velocity vectors at the two positions.

The angle $\Delta\theta$ between the velocity vectors V_1 and V_2 is the same as $\Delta\theta$ in given figures. Since V_1 and V_2 each perpendicular to the radius lines at position A & B respectively, therefore $\triangle ABC$ and $\triangle PQR$ both are isosceles triangles and $\Delta\theta$ are the same.

Hence :

$$\frac{\Delta v}{v} = \frac{\Delta S}{r}$$

Dividing both sides by Δt of the above equation

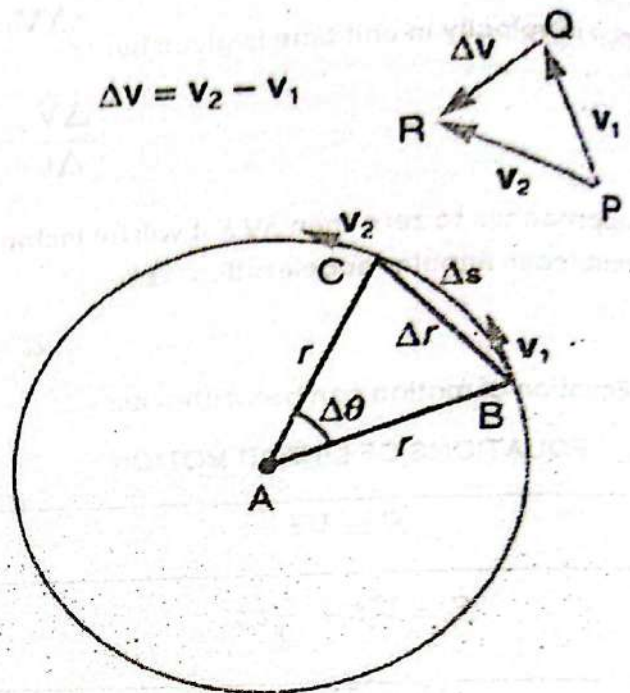
$$\frac{\Delta v}{\Delta t} = \frac{v \Delta S}{r \Delta t}$$

As $\Delta t \rightarrow 0$

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{v}{r} \lim_{\Delta t \rightarrow 0} \frac{\Delta S}{\Delta t}$$

$$a_c = \frac{v}{r} \times v$$

$$a_c = \frac{v^2}{r}$$



In Term Of Angular Velocity

As we know that $V = r\omega$, replace the value of V in above relation

$$a_c = \frac{r^2 \omega^2}{r}$$

$$a_c = r\omega^2$$

Chapter

4

Centripetal Force

CENTRIPETAL FORCE:

"That force which keeps the body in the circular path and acts towards the Centre is known Centripetal Force."

Factors on which the centripetal force depends:

1. Centripetal force is directly proportional to mass of the body.
2. Centripetal force is directly proportional to the square of the velocity.
3. Centripetal force is inversely proportional to the radius of the orbit.

MAGNITUDE OF CENTRIPETAL FORCE

Imagine a ball attached to a string, swinging around in a circle. The ball wants to keep moving in a straight line, but the string pulls it inwards, towards the center of the circle. This pulling force is called centripetal force.

The centripetal force is always directed inwards, towards the center of the circle. It is stronger when the ball is moving faster, and weaker when the ball is moving slower. It is also stronger when the string is shorter, and weaker when the string is longer.

The centripetal force is what makes the ball move in a circular path. If there was no centripetal force, the ball would just fly off in a straight line.

According to Newton's second law of motion we know that

$$F_c = ma_c$$

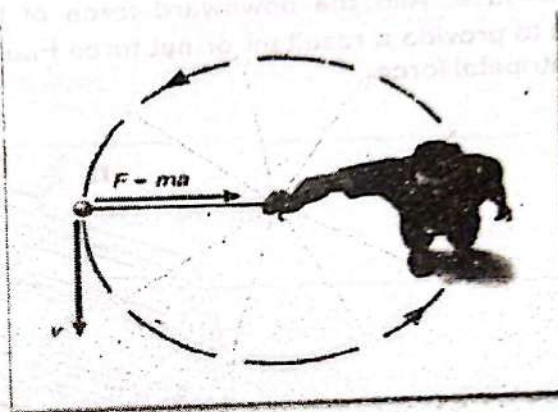
$$F_c = \frac{mv^2}{r}$$

$$[\because v = r\omega]$$

$$F_c = \frac{m(r\omega)^2}{r}$$

$$F_c = \frac{mr^2\omega^2}{r}$$

$$F_c = mr\omega^2$$



Examples Of Centripetal Force

- The force of tension in a string that keeps a ball swinging in a circle
- The gravitational force of the Earth that keeps the Moon in orbit
- The force of friction between a car's tires and the road that keeps the car turning a corner
- The electrostatic attraction between an electron and the nucleus is the centripetal force for the circular motion of the electron around the nucleus.

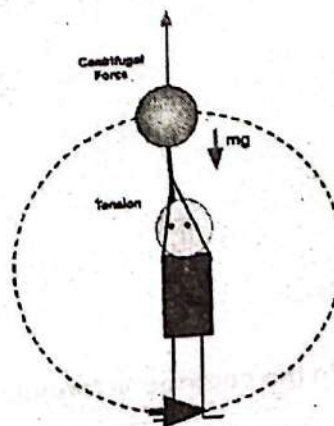
Centripetal Acceleration Caused By Tension Force:

An object under circular motion undergoes centripetal acceleration due to the continuous changes in direction. In cases when the object is tied to a rope, the centripetal force is provided by the tension in the rope. The weight of the body equal to mass $\times$ acceleration due to gravity will be acting in the downward direction.

Mathematically

$$F_c = F_t + F_w$$

$$F_t = \frac{mv^2}{r} - mg$$



Chapter

4

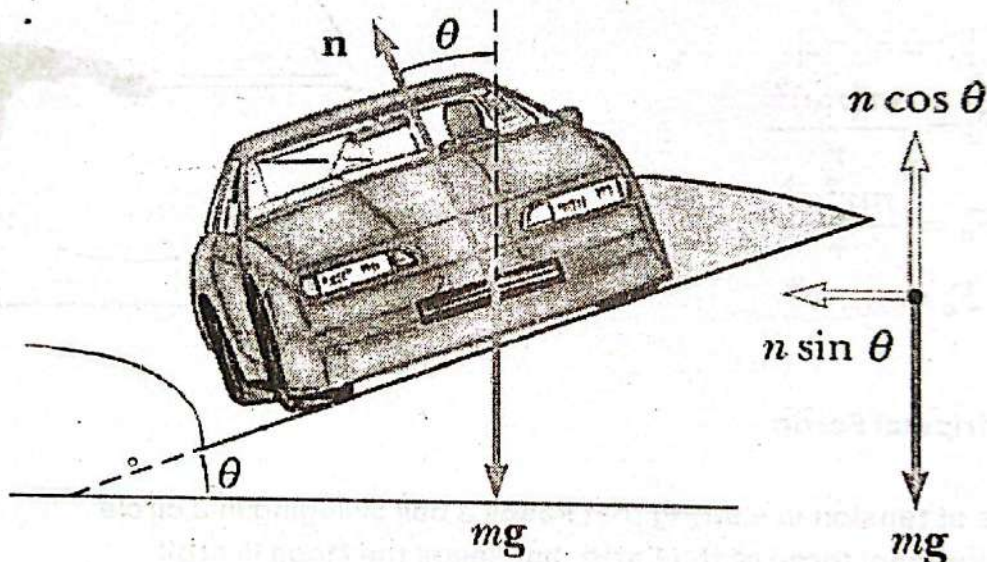
Banked Curve

BANKED CURVE:

A banked curve is a curve in which the road or track is tilted towards the inside of the curve. This tilt creates a force called the banking force, which helps to keep vehicles from skidding. The banking force is equal to the component of the normal force that is perpendicular to the road surface.

Derivation:

A car making a banked turn. Without friction, the roadway still exerts a normal force n perpendicular to its surface. And the downward force of the weight W is present. Those two forces add as vectors to provide a resultant or net force F_{net} which points toward the center of the circle; this is the centripetal force.



Resolve the forces into their components. Since we are interested in the force that points toward the center of the circle, we choose a coordinate axis that lies along that direction. There is no acceleration in the y-direction so the sum of the forces in the y-direction must be zero.

$$F_{\text{net},y} = n \cos \theta - w = 0$$

$$n \cos \theta = w$$

$$n = w / \cos \theta$$

$$F_{\text{net},x} = n \sin \theta$$

$$F_c = mv^2 / r$$

$$F_c = F_{\text{net},x}$$

Substituting in the centripetal formula we get:

$$F_c = mv^2/r = n \sin \theta = [w / \cos \theta] \sin \theta$$

$$F_c = mv^2/r = w [\sin \theta / \cos \theta]$$

$$F_c = mv^2/r = w \tan \theta$$

$$mv^2/r = mg \tan \theta$$

$$\tan \theta = v^2 / gr$$

$$\theta = \tan^{-1} \left(\frac{v^2}{g \cdot r} \right)$$

It shows

- θ is directly proportional to the velocity.
- θ is inversely proportional to the radius of path.
- θ is independent to mass of vehicle.

So, road must be banked for high speed at sharp curves.

Chapter

4

Orbital Velocity

ORBITAL VELOCITY:

Orbital velocity is the velocity at which an object must travel in order to maintain a circular orbit around another object. It is determined by the mass of the object being orbited, the distance between the two objects, and the universal gravitational constant.

Derivation:

let us consider a satellite of mass m revolving around the Earth in a circular orbit of radius r at a height h from the surface of the Earth. Suppose M and R are the mass and radius of the Earth respectively, then $r = R + h$.

To revolve the satellite, a centripetal force is needed which is equal to gravitational force between earth and satellite.

$$F_c = \frac{mv^2}{r}$$

$$F = \frac{GMm}{r^2}$$

By equating both equations we get:

$$\frac{mv^2}{r} = \frac{GMm}{r^2}$$

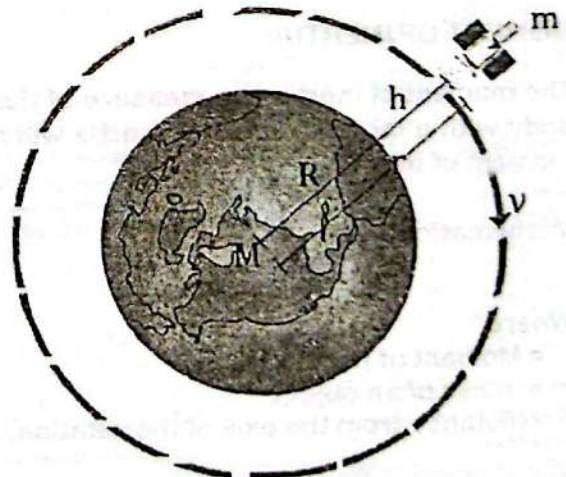
$$v = \sqrt{\frac{GM}{r}}$$

Where,

G is Gravitational Constant

M is Mass of the Earth

r is Radius of the orbit ($R+h$)



ORBITAL TIME PERIOD:

The orbital period is the amount of time it takes a planet or satellite to travel once around its orbit. The linear speed is the distance traveled by the planet or satellite divided by the orbital period. Therefore, the linear speed is equal to the circumference of the orbit divided by the orbital period.

$$v = \frac{2\pi r}{T}$$

From above equation as we know:

$$v = \sqrt{\frac{GM}{r}}$$

$$v^2 = \frac{GM}{r}$$

Equating both equations we get:

$$\left(\frac{2\pi r}{T}\right)^2 = \frac{GM}{r}$$

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

Where:

T = Time period of the orbit (s)

r = Orbital radius (m)

G = Universal Gravitational Constant

M = mass of the object being orbited (kg)

Chapter**4****Moment of Inertia****MOMENT OF INERTIA:**

The moment of inertia is a measure of the resistance of a body to a change in its rotational motion. A body with a large moment of inertia will require more torque to accelerate than a body with a small moment of inertia.

Mathematically:

$$I = mr^2 \quad (\text{For Point Mass only})$$

Where,

I = Moment of Inertia

m = mass of an object

r = distance from the axis of the rotation.

The moment of inertia depends on the following factors:
The density of the material

- Shape and size of the body
- Axis of rotation (distribution of mass relative to the axis)

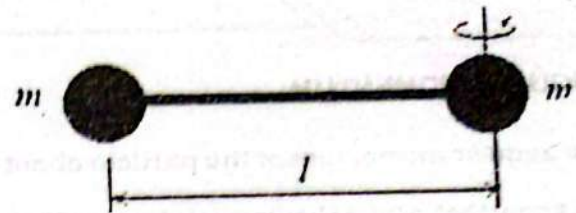
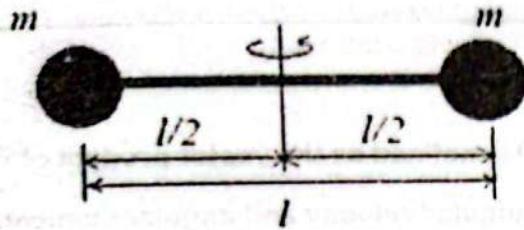
We can further categorize rotating body systems as follows:

1) Discrete (System of particles)

2) Continuous (Rigid body)

MOMENT of INERTIA FOR TWO BODY SYSTEM:

Consider a rigid body containing two particles of masses (m) connected by a rod of length (L) having negligible mass.



The moment of inertia of two particles each at a perpendicular distance of $\frac{1}{2}l$ from the axis of rotation is equal to the product of the mass of each particle and the square of the distance from the axis of rotation, summed for both particles.

$$I = \sum mr^2 = (m)\left(\frac{1}{2}l\right)^2 + (m)\left(\frac{1}{2}l\right)^2$$

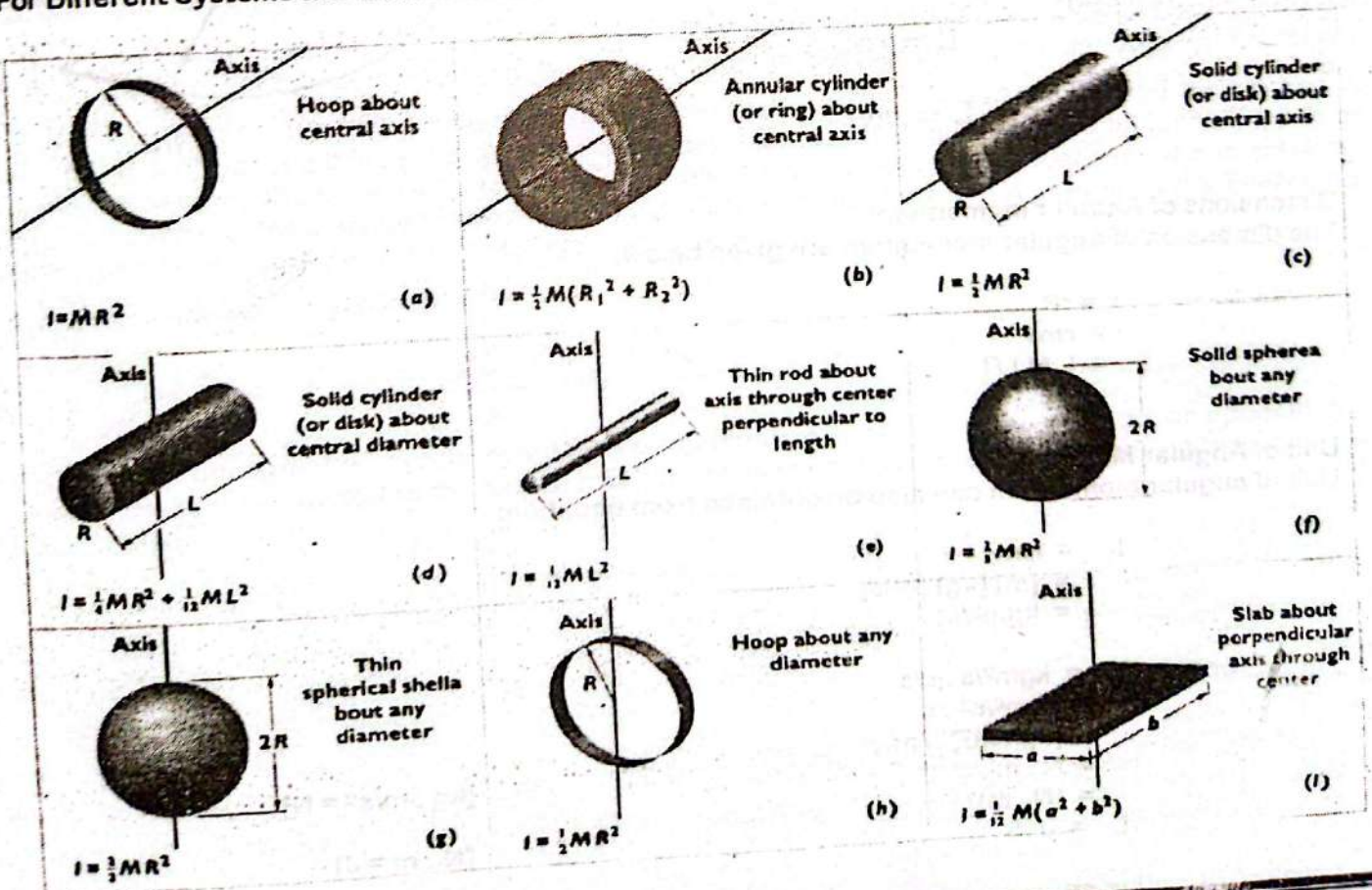
$$I = \frac{1}{2}ml^2$$

Rotational inertia I of the body about an axis through left end of rod and parallel to the first axis. Here it is simple to find I using either method. The first is as in part (a) we have done. The perpendicular distance d is zero for the particle on left and l for the particle on the right. We have:

$$I = m(0)^2 + ml^2$$

$$I = ml^2$$

For Different Systems the Moment of Inertia can be found by different formulae as shown below:



Chapter

4

Angular Momentum

ANGULAR MOMENTUM:

The angular momentum of the particle about the origin O is defined as the vector product of $\vec{r}$ and $\vec{p}$

We know that a body having rotatory motion possesses angular velocity and angular momentum. Angular Momentum like linear momentum obeys the law of conservation. For studying the angular momentum of an object, let us first study the angular momentum of a particle the angular momentum.

Let $\vec{r}$ be the position of a particle of mass 'm' with respect to the origin 'O' shown in the given figure. Moreover let $\vec{p}$ be the linear momentum of the particle measured in an inertial frame of reference with origin O as already shown in the figure.

Hence if ' $\vec{L}$ ' stand for angular momentum then

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\vec{L} = \vec{r} \times m\vec{v}$$

or

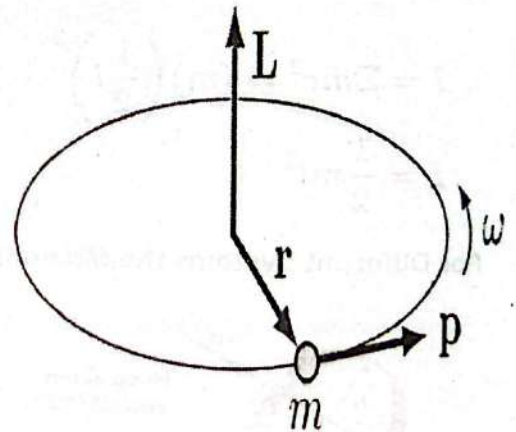
$$L = rP \sin \theta$$

If $\sin \theta = 1$ as $\theta = 90^\circ$

$$L = rp$$

or

$$L = mvr$$



Dimensions of Angular momentum:

The dimension of angular momentum are given below.

$$\begin{aligned} L &= rP \\ &= rmv \\ &= L.M.L/T \\ &= M L^2 T^{-1} \end{aligned}$$

Unit of Angular Momentum:

Unit of angular momentum can also be obtained from equation:

$$\begin{aligned} L &= rmv \\ &= [m] [kg] [m/s] \\ &= kgm^2/s \end{aligned}$$

$$\begin{aligned} \text{or } L &= kgm^2/s \cdot s/s \\ &= kgm^2/s^2 \cdot s \\ &= [kgm/s^2] \cdot (m \cdot s) \\ &= N \cdot m \cdot s \\ &= (N \cdot m) \\ L &= J \cdot s \end{aligned}$$

$$[kg \cdot m/s^2 = N]$$

$$[N \cdot m = J]$$

LAW OF CONSERVATION OF ANGULAR MOMENTUM:

The Law of Conservation of Angular Momentum states that angular momentum remains constant if the net external torque applied on a system is zero.
So, when net external torque is zero on a body, then the net change in the angular momentum of the body is zero.

Derivation:

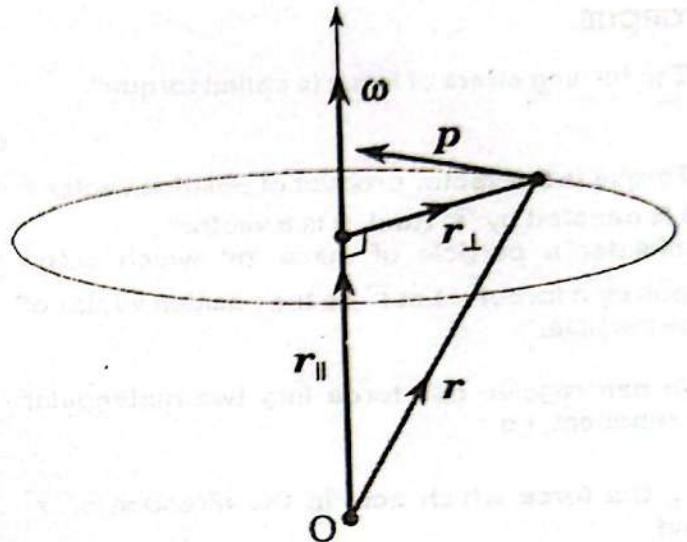
Let us consider a particle in uniform circular motion. The particle is acted upon by a force called the centripetal force, which is directed towards the center of the circle. The centripetal force is necessary to keep the particle in circular motion. The line of action of the centripetal force passes through the center of the circle. Therefore, the centripetal force does not produce any torque about an origin at the center of the circle.

In this case angular momentum will be:

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\vec{L} = (\vec{r}_{\parallel} + \vec{r}_{\perp}) \times \vec{p}$$

$$\vec{L} = (\vec{r}_{\parallel} \times \vec{p}) + (\vec{r}_{\perp} \times \vec{p})$$



The angular momentum of a particle in circular motion has two components: a component along the axis of rotation, which is conserved, and a component perpendicular to the axis of rotation ($L_{\perp} = \vec{r}_{\perp} \times \vec{p}$), which is not conserved. The perpendicular component of the angular momentum is not conserved because it is affected by the centripetal force, which acts perpendicular to the axis of rotation. However, the centripetal force also produces a torque that acts along the axis of rotation, which counteracts the change in the perpendicular component of the angular momentum. Therefore, the total angular momentum of the particle is conserved.

Moreover, because $r_{\parallel}$ is constant we have

$$\frac{dL_{\perp}}{dt} = r_{\parallel} \times \frac{dp}{dt} = r_{\parallel} \times F = \tau_{\perp}$$

so the rate of change of $L_{\perp}$ is supported by the existence of $\tau_{\perp}$. At the same time, the component of angular momentum parallel to the axis of rotation remains constant because there is no torque in that direction.

$$\frac{dL}{dt} = \tau$$

$$\frac{dL}{dt} = 0$$

where $\tau = 0$ (integrating both sides)
 $L = \text{constant}$

Therefore, the angular momentum of a particle is conserved if the net torque acting on it is zero.

Chapter

4

Torque

TORQUE:

"The turning effect of force is called torque".

OR

"Torque is the vector product of position vector $\vec{r}$ and the force $\vec{F}$. It is denoted by ' $\vec{\tau}$ ' (tau). It is a vector".

Consider a particle of mass 'm' which acted upon by a force $\vec{F}$. Let $\vec{r}$ be the position vector of the particle.

We can resolve this force into two-rectangular component, i.e.

$\vec{F}_{\parallel}$, the force which acts in the direction of $\vec{r}$ and

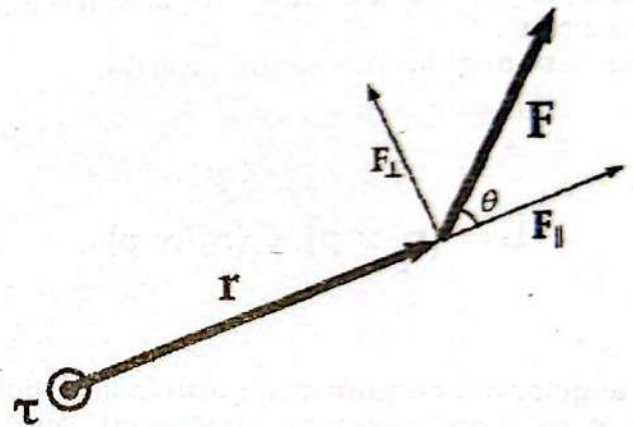
can pull the mass and

$\vec{F}_{\perp}$, the force which acts in the direction Perpendicular to $\vec{r}$ and Produces rotation.

Let 'r' and 'F' be the magnitudes of $\vec{r}$ and $\vec{F}$ respectively. The magnitude of torque $\vec{\tau}$ vector produced by the force $\vec{F}$ about the centre 'O' is expressed as

$$\tau = rF \sin \theta$$

$$\vec{\tau} = \vec{r} \times \vec{F}$$



Direction Of Torque

The direction of torque can also be given by the 'right hand rule'.

Sign Convention

The torque may be clock wise or counter-clock-wise. Hence a torque which produces a counter-clockwise rotation is considered to be positive, while that producing clockwise rotation is taken as negative.

Vector Representation Of Torque

We can represent the torque vector $\vec{\tau}$ in the determinant form, as given below.

$$\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix}$$

RELATION BETWEEN TORQUE MOMENT OF INERTIA AND ANGULAR ACCELERATION

A particle with a mass of "m" is in circular motion at the end of a string with a radius of "r." The mass of the string itself is so insignificant compared to the mass of the particle that it can be disregarded. In this setup, a solitary force denoted as "F" is applied to the mass. This force is responsible for generating a torque that results in an angular acceleration represented by the symbol " α "

As we know :

$$F = m a$$

$$a = r\alpha$$

Substituting we get:

$$\vec{F} = m r \alpha$$

Multiplying both sides by r:

$$r\vec{F} = m r^2 \alpha$$

Where,

$$I = m r^2 \text{ and } \vec{\tau} = \vec{r} \times \vec{F}$$

$$\tau = I \alpha$$

$$\tau \propto \alpha$$

I is called Moment of Inertia which represents rotational inertia

In case of many particle located at various distance from the axis of rotation.

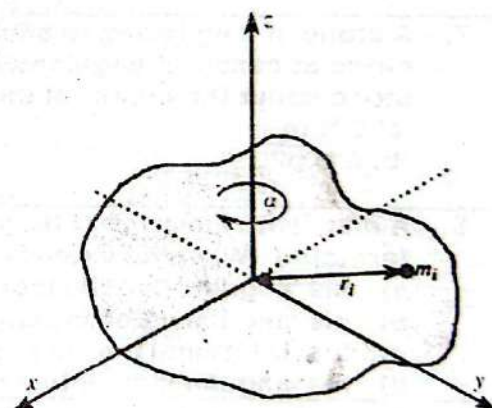
$$\sum \tau = \left(\sum m r^2 \right) \cdot \alpha$$

$$\sum \tau = I \cdot \alpha$$

This is rotational equivalent of Newton's Second law. It is valid for the rotation of a rigid bodies about a fixed axis.

Where

$$I = \sum m r^2 = m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 + m_4 r_4^2 + \dots$$



Chapter

4

Multiple Choice Questions

MCQ'S		ANS.
1. One radian is about a) 25° b) 37°	c) 45° d) 57°	d
2. Wheel turns with constant angular speed then: a) each point on its rim moves with constant velocity b) each point on its rim moves with constant acceleration c) the wheel turns through equal angles in equal times d) the angle through which the wheel turns in each second increases as time goes on		c

3. The rotational inertia of a wheel about its axle does not depend upon its a) Diameter b) mass c) distribution of mass d) speed of rotation	d
4. A force with at given magnitude is to be applied to a wheel. The torque can be maximized by a) applying the force near the axle, radially outward from the axle. b) applying the force near the rim, radially outward. c) applying the force near the axle, parallel to a tangent to the wheel. d) applying the force at the rim, tangent to the rim	d
5. An object rotating about a fixed axis, I is its rotational inertia and α is its angular acceleration. Its a) is the definition of torque b) is the definition of rotational inertia c) is definition of angular acceleration d) follows directly from Newton's second law.	d
6. The angular momentum vector of Earth about its rotation axis, due to its daily rotation is, directed a) tangent to the equator towards east b) tangent to the equator towards the west c) north d) towards the sun	a
7. A stone of 2 kg is tied to a 0.50 m long string and swung around a circle at constant angular velocity of 12 rad/s . The net torque on the stone about the center of the circle is a) 0 N.m b) 6 N.m c) 12 N.m d) 72 N.m	a
8. A man, with his arms at his sides, is spinning on a light frictionless turntable . When he extends his arms a) his angular velocity increases b) his angular velocity remains same c) his rotational inertia decreases d) his angular momentum remains the same.	c
9. A space station revolves around the earth as a satellite, 100 km above the Earth's surface. What is the net force on an astronaut at rest inside the space station? a) equal to her weight on earth b) a little less than her weight on earth c) less than half her weight on earth d) zero (she is weightless)	d
10. If the external torque acting on a body is zero, then its a) angular momentum is zero b) angular momentum is conserved c) angular acceleration is maximum d) rotational motion is maximum	b

***** Exams Practice MCQs *****

1. A body moving along a circular path may have constant (a) Velocity (b) Acceleration (c) Momentum (d) Speed	b
2. The angle between centripetal acceleration and tangential acceleration is (a) 0° (b) 45° (c) 30° (d) 90°	d
3. If "r" is the radius of the circular path of particle, the relation between its linear and angular velocities are. (a) $v = r \times \omega$ (b) $v = \mu \times \omega$ (c) $\omega = v \times r$ (d) $\omega = r \times v$	a
4. One radian is equal to (a) 57.3° (b) 0.017° (c) 67.3° (d) 180°	a
5. The dimension of angular velocity are. (a) T^{-1} (b) LT^{-1} (c) MLT^{-1} (d) LT^{-2}	a
6. A body is moving along a circle with increasing speed it possess (a) Tangential acceleration only. (b) Centripetal acceleration only. (c) No acceleration. (d) Both tangential and centripetal acceleration.	d
7. Every point on the rotating body has the same (a) Linear velocity. (b) Angular velocity. (c) Angular momentum. (d) Torque.	b
8. When a body moves with a constant speed in a circle (a) Its velocity is changing. (b) Its acceleration is zero. (c) Its acceleration is increasing. (d) Its velocity is uniform	a
9. Centripetal force is also called (a) Centre-seeking force. (b) Centrifugal force. (c) Tangential force. (d) Gravitational force.	a
10. When a body moves along circumference of a circle with uniform speed. Changes takes place in its (a) Linear velocity. (b) Tangential acceleration. (c) Radius of circle. (d) Linear velocity and tangential acceleration.	b
11. Angular velocity is also known as angular (a) Rotation. (b) Speed. (c) Frequency. (d) Acceleration.	c
12. A stone weighting 1 kg is tied with 1m long rope and whirled on a circular path with linear speed 1 m/s, tension in the string will be, (a) 9.8N (b) 3.14N (c) 1N (d) 100N	c

Physics XI

13. A cyclist cycling around a circular track skids because: (a) The centripetal acceleration acts upon him is less than the limiting friction. (b) The centripetal acceleration acts upon him is greater than the limiting friction. (c) The centripetal force acts on him is equal to limiting friction. (d) None of these.	b
14. A car is moving on a banked curve at a constant speed. Which of the following forces is the centripetal force? (a) The weight of the car (b) The normal reaction of the road on the car (c) The frictional force between the tires of the car and the road (d) The sum of the normal reaction and the frictional force	c
15. A car is moving on a banked curve at the maximum safe speed. What is the relationship between the angle of banking, θ , and the speed of the car, v ? (a) $\theta = \tan^{-1} (v^2/gr)$ (b) $\theta = \tan^{-1} (v/gr)$ (c) $\theta = \tan^{-1} (gr/v^2)$ (d) $\theta = \tan^{-1} (gr^2/v)$	a
16. A satellite is orbiting the Earth in a circular orbit. If the orbital radius of the satellite is doubled, what will happen to its orbital velocity? (a) The orbital velocity will be halved. (b) The orbital velocity will be quartered. (c) The orbital velocity will remain the same. (d) The orbital velocity will double	a
17. A solid sphere of radius R and mass M is rotating about an axis passing through its center. What is the moment of inertia of the sphere about this axis? (a) $2MR^2/5$ (b) $5MR^2/2$ (c) $7MR^2/5$ (d) $9MR^2/5$	a
18. Which of the following laws states that the sum of the torques on a body is equal to the moment of inertia of the body times its angular acceleration? (a) Rotational equivalent of Newton's second law (b) Newton's second law of motion (c) Law of conservation of angular momentum (d) Law of conservation of linear momentum	a
19. What is the direction of torque? (a) The direction of the torque is perpendicular to the plane of the force & distance. (b) The direction of the torque is the same as the direction of the force. (c) The direction of the torque is the opposite of the direction of the force. (d) The direction of the torque is the direction of the distance.	a
20. What is the law of conservation of angular momentum? (a) The angular momentum remains constant if there is no net torque acting on system. (b) The angular momentum is proportional to the net torque acting on the system. (c) The angular momentum is inversely proportional to the net torque acting on system. (d) The angular momentum is independent of the net torque acting on the system.	a

Chapter

4

Chapter Formulae & Numericals

Angular Displacement

$$\theta = \frac{S}{r}$$

$$1\text{rad} = \frac{180}{2\pi} = 57.3^\circ$$

$$1^\circ = \frac{2\pi}{180} = 0.0175\text{rad}$$

Physics XI

Angular Velocity	$\omega = \frac{\Delta\theta}{\Delta t}$
Angular frequency	$\omega = 2\pi f$ $\omega = \frac{2\pi}{T}$
Angular acceleration	$\alpha = \frac{\Delta\omega}{\Delta t}$
Relation between Linear and angular Velocity	$V = r\omega$
Relation between Linear and angular Acceleration	$a = r\alpha$
Centripetal Acceleration	$a_c = \frac{v^2}{r}$ $a_c = r\omega^2$
Centripetal Force caused by Tension Force in a Vertical Circle.	$F_c = F_t + F_w$ $F_t = \frac{mv^2}{r} - mg$
Bank Curved Angled	$\theta = \tan^{-1}\left(\frac{v^2}{rg}\right)$
Orbital Speed	$V = \sqrt{\frac{G \cdot M_{\text{central}}}{R}}$
Orbital Time period	$T^2 = \frac{4\pi^2 R^3}{GM_{\text{central}}}$
Moment of Inertia	$I = \sum mr^2$
Moment of Inertia for two particle system	Rotation Axis Through Center $I = \frac{1}{2}mL^2$ Rotation Axis through end of the rod $I = mL^2$
Moment of Inertia of Solid Cylinder	$I = \frac{1}{2}MR^2$
Moment of Inertia of Hollow cylinder	$I = \frac{1}{2}m(a^2 + b^2)$
Moment of Inertia of Sphere •	$I = \frac{2}{5}MR^2$